# Math 105L: Using First and Second Derivatives II (Lecture 11-1)

Completed Board Lecture & Discussion Sheet | Monday, November 2, 2026

## 1. Core Principles: Local Extrema vs. Inflection Points

- **Local Extrema (First Derivative Test)**: Let $a$ be in domain of continuous $f$. If $f'(x)$ changes sign at $a$, then $a$ is a local extremum ($+$ to $-$ is a **local max**; $-$ to $+$ is a **local min**). A critical point ($f' = 0$ or undefined) is *necessary* but *not sufficient*.
- **Points of Inflection**: A point $(c, f(c))$ is an **inflection point** if $f$ is continuous and the **concavity changes sign** across $c$. The condition $f''(c) = 0$ (or undefined) is a candidate requirement, but concavity must strictly alternate ($f''$ changes sign).
- **Critical Warning (Common Exam Pitfall)**: Setting $f''(x) = 0$ does **NOT** find local extrema! Local extrema occur where $f'(x) = 0$ (or undefined) and $f'$ changes sign. The second derivative test classifies an *already identified* critical point ($f'(a) = 0$) by checking the sign of $f''(a)$.

| **Function** | $f'(x)$ | $f''(x)$ | **Candidate at** $x = 0$ | **True Classification at** $x = 0$ |
|---|---|---|---|---|
| $y = x^3$ | $3x^2 \geq 0$ | $6x$ | $f'(0) = 0, f''(0) = 0$ | **Inflection Point** ($f''$ changes $-$ to $+$); *Neither* max nor min. |
| $y = x^4$ | $4x^3$ | $12x^2 \geq 0$ | $f'(0) = 0, f''(0) = 0$ | **Local Minimum** ($f'$ changes $-$ to $+$); *NOT* an inflection point ($f'' \geq 0$). |
| $y = -x^4$ | $-4x^3$ | $-12x^2 \leq 0$ | $f'(0) = 0, f''(0) = 0$ | **Local Maximum** ($f'$ changes $+$ to $-$); *NOT* an inflection point ($f'' \leq 0$). |

## 2. Case Study: The High-Degree Polynomial $f(x) = (x^2 - 4)^7$

**Algebraic Properties**: Degree $2 \times 7 = 14$ (even degree, positive leading coefficient $\implies \lim_{x \to \pm\infty} f(x) = +\infty$). Roots at $x = \pm 2$.

**First Derivative & Critical Points**: $f'(x) = 7(x^2 - 4)^6 \cdot (2x) = 14x(x^2 - 4)^6$. Setting $f'(x) = 0 \implies 14x = 0$ or $(x^2 - 4)^6 = 0 \implies \mathbf{x = 0, \pm 2}$. Since $(x^2 - 4)^6 \geq 0$ for all $x$, the sign of $f'(x)$ is governed strictly by $14x$:

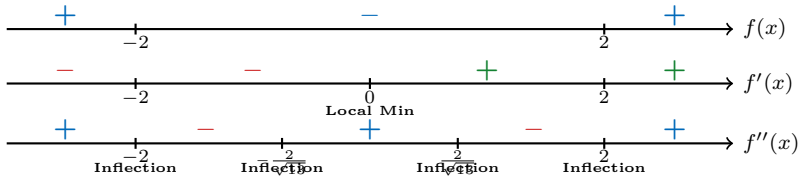
- $x < 0 \implies f'(x) < 0$ (**decreasing**); $x > 0 \implies f'(x) > 0$ (**increasing**).
- At $x = 0$: $f'$ changes from $-$ to $+ \implies$ **Local Minimum** at $(0, -16384)$ where $(-4)^7 = -16384$.
- At $x = \pm 2$: $f'$ does **not** change sign ($f' \leq 0$ on both sides of $-2$; $f' \geq 0$ on both sides of $2$) $\implies$ **Neither max nor min!**

**Second Derivative & The Factoring Strategy**: Applying product rule and chain rule to $f'(x) = 14x(x^2 - 4)^6$:

$$f''(x) = [14](x^2 - 4)^6 + 14x \cdot \left[6(x^2 - 4)^5 \cdot (2x)\right] = 14(x^2 - 4)^5 \left[(x^2 - 4) + 12x^2\right] = \mathbf{14(x^2 - 4)^5(13x^2 - 4)}$$

Setting $f''(x) = 0$ yields 4 candidate inflection points: $x^2 - 4 = 0 \implies x = \pm 2$ and $13x^2 - 4 = 0 \implies x^2 = \frac{4}{13} \implies x = \pm\frac{2}{\sqrt{13}} \approx \pm 0.5547$.

**Non-Calculator Sign Analysis via Parabolas**: $(x^2 - 4)^5$ has the same sign as $x^2 - 4$ (negative inside $(-2, 2)$, positive outside). $13x^2 - 4$ is a parabola opening up (negative inside $(-\frac{2}{\sqrt{13}}, \frac{2}{\sqrt{13}})$, positive outside).




**Conclusion**: All four candidate roots ($x = -2, -\frac{2}{\sqrt{13}}, \frac{2}{\sqrt{13}}, 2$) exhibit sign changes in $f''(x)$. Therefore, **all four points are true inflection points!**